Digital System Design

Midterm(97/04/07)

Spring 2008

Total points: 105points

1. A sequential circuit with two D flip-flops, A and B; two inputs, x and y; and one output, z, is specified by the following next-state and output equations:

$$A(t+1)=x'B+xy$$

$$B(t+1)=x'A$$

$$Z=AB$$

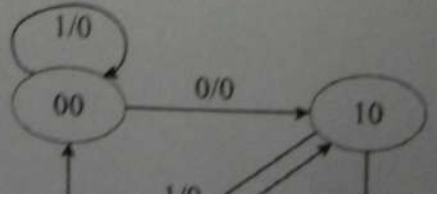
(a) Draw the logic diagram of the circuit. (5%)

(b) List the state table for the sequential circuit. (15%)

2. A state diagram with two state variables D_A and D_B and one output z is shown in the following figure,

(a) List the state table. (5%)

(b) Write down the maps of the two state variables D_A and D_B and the output z; and then derive the Boolean equations for the three maps. (15%)



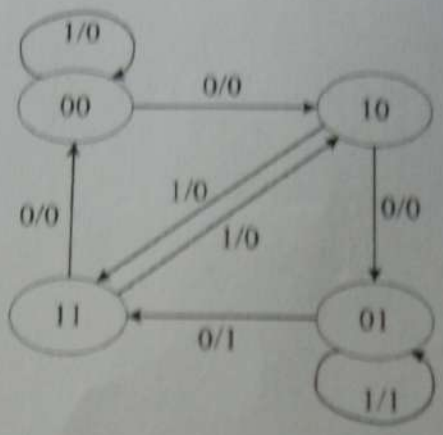
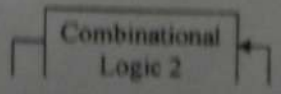


Fig. 1 State diagram

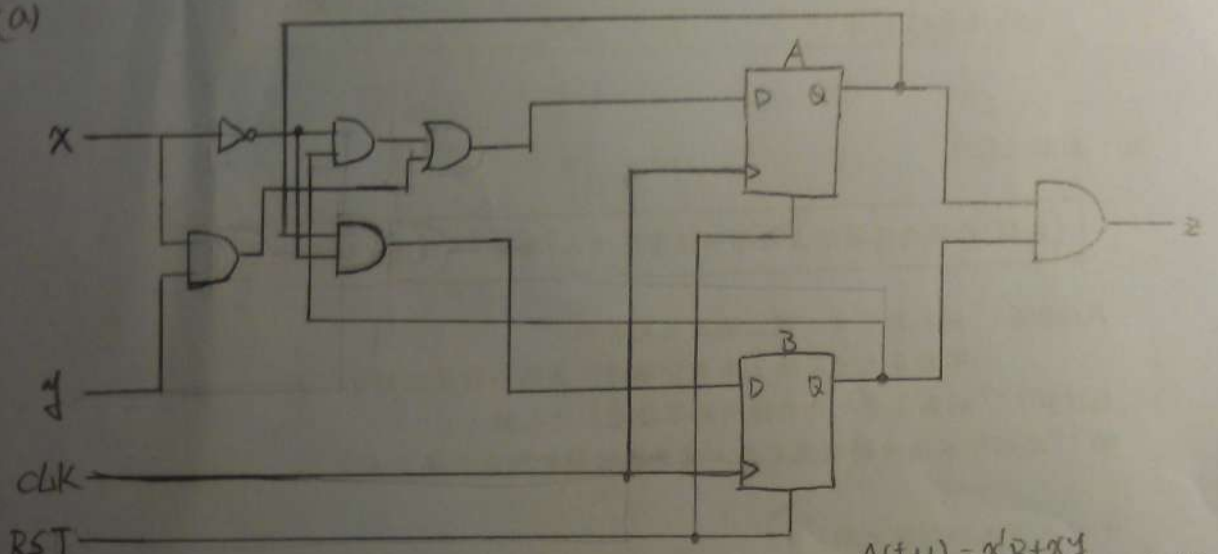
3. Given the circuit connected as shown in Fig. 2, assume that the wire delay is **not** considered and the D flip-flop has setup time of 2 ns and hold time of 1 ns.
   (a) If the gate delay of the "**combinational logic 1**" is 12 ns and the gate delay of the "**combinational logic 2**" is 5 ns, what is the minimum clock period of this circuit to have normal operation? (5%)
   (b) If "**combinational logic 1**" and "**combinational logic 2**" are both replaced by the same new logic block with minimum delay requirement, what is the minimum clock period of the new circuit to have normal operation? (5%)



zoom 8

1.

(a)



$A(t+1) = x'B + xy$

$B(t+1) = x'A$

$z = AB$

(b)

Present state	Input	Next state	Output

b)

Present state		Input		Next
A	B	x	y	A
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

$B(t+1) = x'A$

Next state		Output
A	B	z
0	0	0
0	0	0
0	0	0
1	0	0
1	0	0
1	0	0
0	0	0
1	0	0
0	1	0
0	1	0
0	0	0
1	0	0
1	1	1
1	1	1
0	0	0
1	0	0